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EXTENDED-ABSTRACT

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Autonomous agents in software development for information retrieval using LLM models

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Abstract

Currently, the Large Language Models (LLMs) and autonomous agents are rapidly evolving technologies with the potential to revolutionize a variety of industries. The paper examines the application of autonomous agents in information retrieval using LLMs. Its purpose is to investigate how these technologies can enhance the efficiency, speed, and reliability of fact-finding and question-answering. Paper describes the development idea of information retrieval system utilizing autonomous agents and selected LLMs.

CCS Concepts

• **Computing methodologies → Natural language processing**; • **Software and its engineering → Software organization and properties**.

Keywords

LLM models, autonomous agents, information retrieval, natural language processing, question answering

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1 Introduction

Large Language Models (LLMs) and autonomous agents are rapidly evolving technologies with the potential to revolutionize a variety of industries. The advanced capabilities of LLMs allow them to understand and generate human-like text, making them invaluable for tasks such as content creation, decision-making, and data analysis. The ability of LLMs and autonomous agents to perform complex tasks, facilitate communication, and automate decision-making without human intervention is one of their key strengths. This ability to function without constant human supervision can lead to increased efficiency and innovation in a variety of sectors. However, while LLMs and autonomous agents offer significant advances in automation and intelligence, the complexity of developing and integrating these systems poses a challenge for developers seeking to fully leverage their capabilities. Autonomous agents are characterized by several key features that enable their independent action, intelligent behavior, interaction: autonomy, reactivity, proactivity, social capabilities, learning and adaptation, mobility [1–3].

Large Language Models and autonomous agents pose a number of challenges, particularly in the areas of knowledge management and the reliability of the information they provide. Developing and implementing these technologies requires developers to have a deep understanding of natural language processing, machine learning algorithms, and data management. LLMs often struggle to provide consistently accurate and reliable information, as they can generate plausible-sounding but incorrect or misleading answers. This problem is compounded by the fact that LLMs may not have access to up-to-date information in real time, leading to outdated or contextually inappropriate results [4–6].

Paper presents the idea of using the selected large language models in autonomous agent systems, focused on the implementation of information retrieval tasks. Our study focused on a detailed evaluation of selected LLMs and critical review of existing solutions.

2 Information retrieval using multi-agent systems and LLMs

The development of proposed system was primarily motivated by the lack of a reliable, open and cost-effective solution for comprehensive information retrieval. In response to this gap, an attempt was made to create an innovative platform that combines innovative features, distinguishing it from other currently existing multi-agent systems. The proposed method was designed to excel in fast information retrieval and synthesis in a wide range of topics, from daily news to in-depth academic research. Its basic function is based on an advanced process that includes question understanding, planning, finding helpful facts and detailed information analysis, report generation and rigorous quality control at each stage of the process aimed at finding an answer to the user's question. The overarching goal of this system is to provide highly reliable and trustworthy information, establishing a new approach in automated research processes. The architecture of proposed solution is conceptually organized into 4 main modules, each representing a separate phase in information retrieval and synthesis process. Modules can be as specialized teams of agents within larger, coherent system (Fig. 1).

*Both authors contributed equally to this research.

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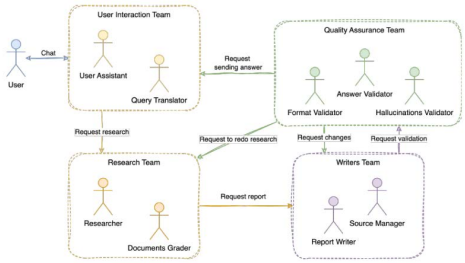

Figure 1: Agent modules cooperating within a multi-agent system and their inter-team communication.

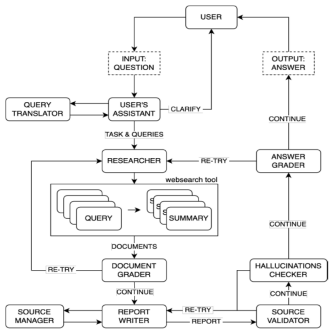

Figure 2: Workflow schema of LLM-based agent system.

The system is designed to provide comprehensive and authoritative answers to user queries, while rigorously adhering to citation guidelines. The entire response process is presented in figure 2. The process begins with a thorough analysis of the user's query to ensure a complete understanding of the user's intent. This initial step is particularly important for overly general questions that require refinement to achieve optimal question precision. For example, a general question such as "What are the effects of climate change?" can be refined to "What are the effects of climate change on coastal ecosystems in the Mediterranean over the past decade?". After an initial task understanding phase, the system generates a series of targeted search queries. These queries are designed to explore a variety of online resources, including academic databases, recent press releases, and trustworthy websites. This comprehensive search strategy ensures that a broad spectrum of information is collected, covering both academic research and breaking news.

Then, another agent analyzes and evaluates the relevance and credibility of the collected information. If the collected data does not meet predefined quality standards, the system initiates a recursive research process to obtain more relevant information. On the other hand, if the collected data is deemed good enough and reliable, it is forwarded to specialized agents responsible for writing reports and managing sources. These agents synthesize the information into a coherent and structured response that directly addresses the user's query. The generated response then undergoes a rigorous verification process. This complex verification includes checking the accuracy of sources, detecting potential hallucinations or unsupported claims, and assessing the overall relevance and usefulness of the response with respect to the original question. Only after these verification procedures have been successfully completed is the final response presented to the user via an intuitive and user-friendly interface. This designed process ensures the delivery of high-quality, well-documented and directly related to the user's query responses, maintaining the system's commitment to the accuracy and credibility of the information disseminated.

Figure 2 presents the complete process of proposed multi-agent solution with detailed description of each agent, its responsibilities, and communication process between agents. Each of the listed agents plays only a specific role within the system. The *User Assistant* serves as a key interface between the user and the system, performing a number of important functions. Its primary task is to evaluate the specificity and clarity of the user's initial query. The *Context Translator* plays a key role in the information retrieval process, serving as a bridge between the user's query and the query-structured topics related to the user's intentions. The *Researcher* occupies a central and key place in the system's process architecture, responsible for conducting comprehensive and detailed information retrieval. The *Content Verifier* plays a key decision-making role in the system's workflow, acting as an evaluation filter for the retrieved information. During the information generation process, two closely cooperating agents play a key role: the Report Author and the Source Manager. The *Report Author* serves as the main information synthesizer, responsible for creating an accurate and comprehensive answer to the user's query. The *Source Manager*, working in synergy with the Report Author, plays a specialist role in the process of verifying and formatting citations.

In the process of selecting appropriate large language models for the proposed system, a comprehensive evaluation of available models based on several key criteria was carried out. The analysis of benchmark results, such as LLMU, GPQA, DROP, MGSM, MATH, HumanEval, MMMU, revealed that individual LLM models show different levels of proficiency in different types of tasks. This required a thorough process of validation of effectiveness of individual LLMs in order to identify the models most suitable for the specific requirements of the proposed solution. The following models were selected for detailed evaluation: Llama2, Llama3, Gemma2.

3 Conclusions

The paper presented approach to automating knowledge acquisition by developing and building a system using autonomous agents with LLMs as their cognitive and operational cores. Created framework represents a step forward in the field of information retrieval.

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